

Machine Learning-Based Sitting Posture Recognition for Prolonged Use of Technology in Office Environments

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Alpas, John Rafael D.

School Of Information Technology Mapúa University,

jrdalpas@mymail.mapua.edu.ph

Reyes, Elijah Erle O.

School Of Information Technology, Mapúa University,

eeoreyes@mymail.mapua.edu.ph

Taningco, Kurt Andrei C.

School Of Information Technology, Mapúa University,

kactaningco@mymail.mapua.edu.ph

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Chapter 1

1.1. Introduction

Although a hive of activity, the modern workplace seriously compromises the physical wellbeing of its employees. A feature of modern office environments, the prolonged durations of sitting involvement clearly help to produce postural abnormalities. This has then caused a rise in musculoskeletal conditions like general muscular tiredness, spinal misalignment, and chronic back discomfort, affecting individual health as well as organizational effectiveness.

Algorithms such as MediaPipe and YOLOv5 enable post-analysis of seating positions. The advanced models, trained on vast amounts of datasets, present a viable avenue for the diagnosis and correction of improper postures in the setting of an office environment. Though they display technological capabilities, it is highlighted that contemporary solutions lack the necessary flexibility for effective assimilation into the bustling office environment. Moreover, one still has much work to do on the provision of immediate, helpful feedback. Current solutions fail to fulfill the complex ergonomic needs of the workplace and primarily concentrate on generalized posture monitoring. Therefore, the development of a sophisticated posture detection algorithm capable of delivering remedial help is plainly important in order to lessen the health concerns related to extended sedentary behavior in a way that is both efficient and inconspicuous.

1.2. Background of Study

The increasing use of technology in office environments and other fields has led to prolonged sitting, often resulting in poor posture or ergonomics, which is a risk factor that can lead to various related musculoskeletal disorders (MSDs) like back pain [1, 2]. Addressing this

issue, recent advancements in analyzing human motion and posture recognition algorithms leverage machine learning and sensor-based technologies to monitor and correct sitting postures. Studies on human activity recognition, machine learning algorithms for pose estimation, force sensors, and inertial measurement units (IMUs) highlight the effectiveness of these techniques in detecting and analyzing posture. Machine learning models have proven to be crucial in recognizing improper sitting postures and helping to prevent health risks associated with extended periods of technology use [3]. Our posture while sitting, for example, can cause people health problems that may affect our work and cause us to perform poorly, even to the point of needing to ask for sick leave [4]. Health-related studies say poor body posture may lead to back pain, organ displacement, poor balance, breathing difficulties, and bone degeneration [5]. Data shows that most office workers spend more than 4 hours per day working, often working in front of a computer, which indicates that they are sitting in front of a computer for a long time, which could result in poor body posture [6]. It is likely to happen and become unnoticeable due to the workers being busy, and their posture will likely deteriorate the longer they spend working in their workstations. This gives us more reason to pay more attention to this problem and create solutions that may help combat it.

Computer vision, a field of artificial intelligence, empowers machines to interpret and understand visual information from the real world, much like human vision. Computer vision mimics the human visual system, where our eyes are responsible for sensing our environment. At the same time, the brain is responsible for understanding it; computer vision algorithms can analyze images and videos with the use of cameras as their sensing device and gadgets like computers to act as the brain to interpret what the camera sees, extracting meaningful insights and data [7]. Computer vision could help with problems such as poor body posture by recognizing different postures visually because of its capability to see visual information with the use of cameras, in which it could detect the key points to identify if the posture is poor or bad [8].

Machine learning is used by computer vision to gather data to allow it to see visually because of its trained models and a large dataset of images and video, which makes machine learning a crucial part of computer vision to enable it to perform well [9]. The issues of computer vision, for example, like feature extractions and processing, are resolved with the help of machine learning, which further solidifies its importance for computer vision [10]. Much research is still being done about computer vision, but it has a significant value in medical, biology, military, and other fields [11]. Computer vision is used by different frameworks or algorithms such as MediaPipe which is a framework that has a practical pose estimation that is popular among fitness fields because of its ability to identify and capture essential anatomical landmarks which is effective to analyzing body movement during exercise, YOLOv5 is also a detection system model that is used in computer vision which makes it a good pair with Mediapipe [12, 13]. Using both MediaPipe and YOLOv5 will be effective in recognizing body posture to identify poor body posture and avoid health issues.

Posture has a prevalent impact within prolonged desk-bound working environments, where deterioration of posture because of prolonged technology use may cause different kinds of pain, like musculoskeletal-related pain, that could hinder work performance and cause detrimental health issues in office workers [14]. International standard ISO 11226:2000 identifies five basic anatomical elements of posture that are critical to the ergonomics of the workplace: the trunk, head, shoulder, and forearm/hand [15]. The elements provided by ISO 11226:2000 possess a procedure that will evaluate the workers' static posture and determine whether they are healthy or unhealthy. The identification of these basic elements of posture is important for building effective recognition systems because they set the foundation for the detection and correction of postural errors. Poor postural behaviors in our adulthood are often formed in childhood, which is worsened in the workplace, where workers tend to spend 4 hours per day sitting due to work [6, 16]. Poor posture that is tolerated will result in decreased cardiorespiratory fitness, decreased

lung capacity, and chronic low back pain. These health consequences may significantly affect office productivity and thereby impact long-term occupational health.

Computer-centric work models are a dominating onsite environment in the Philippines where high density and technology-driven workplace dynamics are far reaching, profoundly shaping occupational health. As an economic hub, Metro Manila accommodates over 1.2 million Business Process Outsourcing (BPO) workers, enduring prolonged shifts through repetitive computer use with non-ergonomic setups. Latest research indicates that 76% of Filipino BPO workers report chronic back pain, with 63% experiencing neck strain, presenting high posture-related injuries among office workers within the BPO sector [17]. The BPO sector contains a standardized workstation layout, while uniform, introduces poorly designed ergonomics. This challenge underscores systematic analysis for ergonomic flaws, an important sector for conducting recognition research in highly computer reliant environments.

Figure 1 highlights the reported MSDs of both BPO and work-from-home employees, presenting valuable percentages across different body areas. A recent study focusing on the BPO population reported disproportionate findings. The BPO pain rates were significantly higher with the neck (51%) and hip (51%) pain areas being comparatively higher than the work-from-home neck (11%) and hip (5%) rates[18]. Moreover, ergonomic assessments conducted locally within the Philippines presented similar findings, with a study of onsite workers having higher lower back pain (46%) and wrist pain (44%) being most affected. The contrast observed within these workplaces addresses the limiting adjustability of BPO workstations, which offer minimal ergonomic solutions compared to work-from-home setups, subsequently contributing to a higher prevalence of MSDs [19, 20].

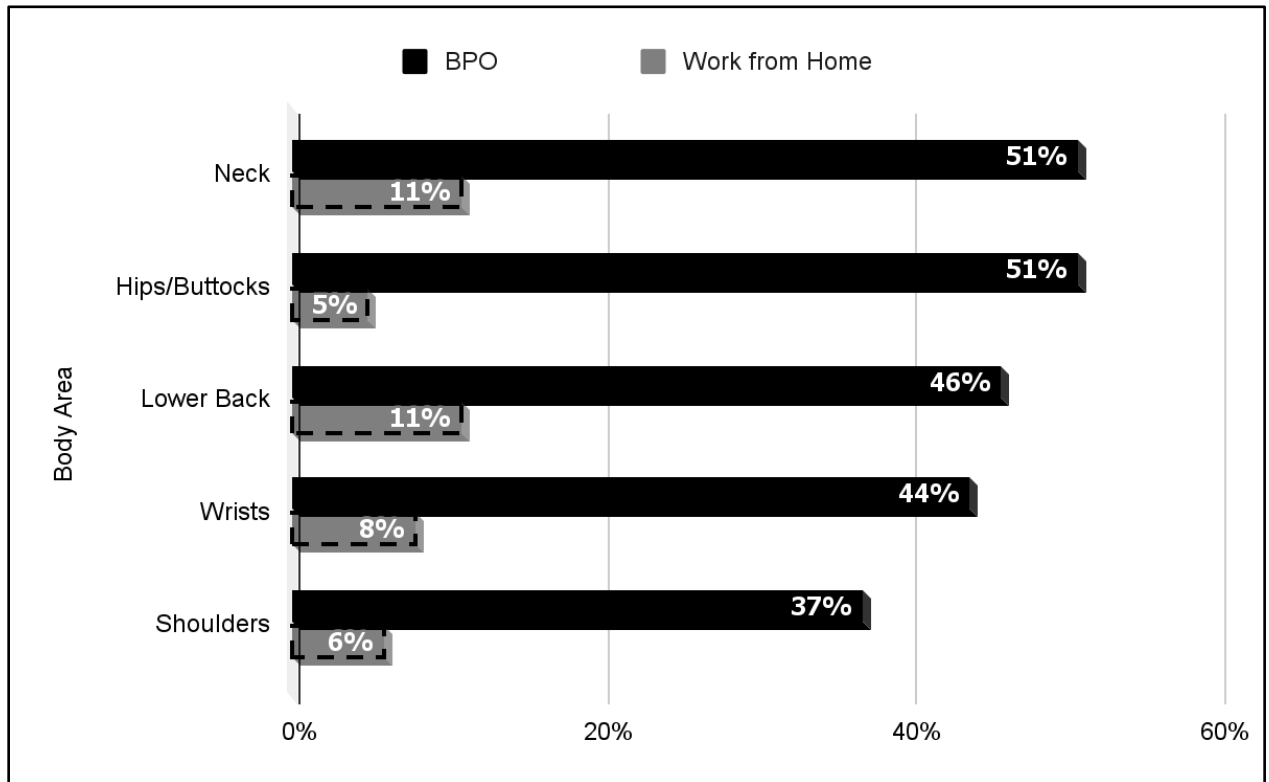


Figure 1. Pain Area Prevalence

Work-related musculoskeletal disorders pose a notable challenge to occupational health in the Philippines. In 2015, around 79,000 cases related to MSDs were documented, and this trend appears to be increasing [21]. Chronic musculoskeletal diseases have become prevalent in the workplace and often result from poor posture, overuse injuries, and long hours of sitting. Ergonomic hazards, such as back pain, muscular fatigue, and repetitive stress injuries (RSIs), have an extensive effect on the wellness of workers in these environments [22]. Considering that 51% of businesses are now using onsite workers again, according to a survey from 2023, this trend is only going to accelerate [23]. Improved working conditions that promote health and wellness are more important than ever before. It is intriguing to further delve into the potential of developing a more secure and efficient workplace. By mitigating ergonomic risks with

innovative solutions such as posture recognition technology, taking substantial steps in alleviating musculoskeletal disorders.

1.3. Gap Analysis

Despite the existing advancements in monitoring an individual's posture, there remains a deficiency of accessible posture recognition algorithms that can be implemented, particularly within office environment. Most current solutions focus solely on general posture monitoring, lack actionable feedback for behavioral change, or do not offer sufficient adaptability for dynamic office settings, where workers may not have the time or awareness to self-correct their posture. This gap underscores the necessity for an intuitive algorithm that not only detects improper posture but also provides users with corrective feedback to mitigate the cumulative health effects of prolonged poor sitting habits. While numerous studies recognize the increasing health risks associated with poor posture, limited attention has been given to the application of corrective technology in office settings within the Philippines. Addressing this gap involves developing a posture recognition algorithm that can be practically implemented to enhance workplace wellness.

The MediaPipe Hands solution shows how on-device tracking can be done with a simplified and effective two-stage pipeline that can guess 2.5D hand landmarks from a single RGB camera on several different devices. Although the algorithm exhibits robust tracking capabilities for AR/VR applications, it lacks the capability of immediate adaptive feedback to correct user posture. This deficiency represents a significant challenge for posture recognition algorithms employed in workplace ergonomics overall. It has been shown in new studies on biomechanical posture analysis that machine learning (ML) models can accurately and reliably judge posture, with intraclass correlation coefficient (ICC) values of up to 0.95 for individual posture measurements. The demand for adaptable solutions capable of addressing the dynamic

characteristics of office environments is underscored by the observation that these methods are generally limited to regulated, static situations. Recent improvements to Mediapipe's three-dimensional posture detection show how hard it is to keep things stable and accurate in-depth when people are moving quickly. These improvements show how jitter and latency make it impossible to correct things in office environments. Considering previous shortcomings, these studies emphasize the essential requirement for a posture recognition algorithm capable of adapting to the dynamic conditions of a hectic work environment while providing immediate feedback.

Posture recognition systems, although arriving at advanced iterations, exhibit critical limitations. Modern solutions possess a reliance on static datasets or single-frame analysis, by extension removing the adaptability that dynamic office environments require. This versatility necessitates adaptive corrective solutions, a unique challenge that MediaPipe and YOLOv5 aims to investigate within an underexplored ergonomic discrepancy. Such mismatches are commonplace within the arrangement of worker setups found in office settings. The orientation of furniture, monitors, and computer accessories underlines inconsistent work environments that disregard ergonomics [24]. A posture recognition model must adapt to these variables for effective posture feedback, addressing specific requirements that can range from environmental to physical.

Figure 2 shows the features that the study's model aims to achieve and its comparison to an already existing model that is sourced from research about a computer vision based human body posture detection system [8]. It shows that both models integrate the key features of pose detection. However, the focus and additional functions of the study's model is different from the existing model that is being compared to.

Features	Proposed Model	Yangsen Qiu et al. Model
Anatomical Key-point Tracking	✓	✓
Real-Time detection	✓	✓
Real-Time & Post Analysis	✓	X
Office Focused	✓	X

Figure 2. Model Feature Comparison

1.4. Research Question

Different variables impact the rate of postural deterioration across an extended workday, many of which correspond to the effectiveness and accuracy of pre-existing training models. As prolonged desk-bound work persists.

- What is the impact of extended computer use on gradual shifts in posture, and to what extent can improper posture lead to noticeable discomfort throughout a workday?
- To what degree can the MediaPipe or YOLOv5 algorithm effectively identify and deliver feedback on improper seating postures in an office setting?

1.5. Research Objectives

This study develops and evaluates a Tensorflow and MediaPipe-based algorithm combined with YOLOv5 for posture identification to monitor office workers' posture. The focus is on creating a reliable tool for detecting improper sitting postures and providing corrective feedback, emphasizing the detection accuracy and responsiveness of the algorithm. Observing changes in posture patterns and reductions in discomfort over time will assess the algorithm's contribution to users' posture-related health. Additionally, the study aims to contribute to ergonomics by examining the benefits and drawbacks of focusing on a single approach to address poor posture in office environments.

- Develop a posture recognition model with the use of MediaPipe, Tensorflow and YOLOv5 (which follows the same technique of another literature) for pose detection with the help of machine learning to recognize bad postures in an office environment.
- The model will detect and analyze the improper sitting posture of office workers and provide corrective feedback using data and graphical representations while also comparing it to other similar models from the internet.

1.6. Scope and Delimitation

The proposed recognition algorithm will use MediaPipe, Tensorflow and YOLOv5 to create an algorithm that will detect the body postures of people. MediaPipe already has multiple integrated algorithms behind it because it has access to TensorFlow which has a lot of tools and algorithms to help build the study's model. YOLOv5 will be used because another study adopts the same concept of using both MediaPipe and YOLOv5 which will be a good reference for making the model. Roboflow universe will also be used to obtain the trained data that will determine improper and proper posture by incorporating 500-1000 labeled images from the dataset available. The estimated epochs that will be used in training will be around 100 and

above. The algorithm aims to recognize body posture of a single person sitting in a side-view perspective and tell if the posture being detected is improper and provide corrective feedback to avoid health-related issues. The algorithm's goal is only to acknowledge body postures both good and bad and will not have any other functions unrelated to detecting body posture.

1.7. Significance of the Study

This study aims to contribute to a healthy lifestyle for workers while in the office by promoting proper posture, which is essential for overall wellbeing and productivity. Maintaining good posture can help reduce the risk of musculoskeletal disorders, chronic pain, and other health issues caused by prolonged sitting and poor ergonomic practices. By raising awareness about the importance of proper posture, the study seeks to educate office workers and the general public on the long-term benefits of maintaining a healthy posture, which is often overlooked in daily routines. Additionally, this study highlights the risks associated with poor body posture, such as back pain, neck strain, and fatigue, which can negatively impact work performance and quality of life. Beyond its direct impact on office workers, this study also contributes to the advancement of body recognition technology, which is valuable for computer scientists working on the development of programs or algorithms related to posture correction and health monitoring.

By exploring ways to improve posture recognition, the findings of this study can be applied in fields such as artificial intelligence, machine learning, and human-computer interaction. This research may serve as a foundation for the development of innovative solutions, such as smart posture-monitoring systems, wearable devices, or AI-powered ergonomic assessment tools that can benefit individuals and organizations. By bridging the gap between technology, health, and workplace ergonomics, this study provides valuable knowledge that can be utilized in various disciplines. It not only addresses an often-overlooked health concern but

also fosters innovation in body recognition technology, benefiting a wide range of industries and individuals

Chapter 2

2.1. Introduction

Focusing on recent advancements in machine learning and sensor technology. It throws light on the methods, algorithms, and tools that are being used in monitoring postures, in the contexts of working in the office and health-related contexts, where the issue of how such approaches may mitigate prolonged poor posture-induced musculoskeletal issues is raised. The reviewed studies lay down a foundation to understand the context created by the current technologies and methodologies regarding algorithms for posture recognition.

2.2. Review of Related Literature

2.2.1 Technological Advancements to Prevent Musculoskeletal Disorders

Musculoskeletal disorders can affect muscles, bones, and are a prominent risk factor that could occur due to poor body posture. Spinal discs and back muscles can be strained due to bad posture, other health issues could also occur like poor circulation, diminished lung function, trunk asymmetry, and others. Advancements made in technology opened opportunities to help prevent these risk factors. Specifically, motion and posture analysis are rapidly progressing which allows us to detect and track poor body posture which can help with the prevention of musculoskeletal disorders and other issues related to bad body posture. This is done by identifying anatomical landmarks by either having trained professionals identify it or use a trained algorithm to identify anatomical landmarks. The anatomical landmarks will then be

detected by hardware like an optical scanner to detect body posture or inertial measurement units that are being developed due to technological advancements [1].

2.2.2 The Role of Machine Learning in Posture Prediction Models

Machine learning models have been employed to predict posture trends over time, allowing early intervention before musculoskeletal issues arise. By training recurrent neural networks (RNNs) and long short-term memory (LSTM) models using historical posture data, researchers have developed predictive models that can forecast future postural deviations based on user habits. TensorFlow has been instrumental in developing such models, enabling personalized posture recommendations tailored to an individual's movement patterns. Studies show that predictive analytics in posture monitoring can significantly reduce the likelihood of chronic pain and musculoskeletal complications.

To properly handle posture data, CNNs assess video and photographic feedback by visually scanning for posture variations through the assistance of LSTMs. Spatial information such as spinal curvature and shoulder alignment are important factors that LSTMs track over time. Hybrid models such as these combine input from a number of devices, more notably motion trackers or sensors, and clean footage by utilizing noise reduction and time-series normalization techniques. Autonomous adjustments throughout various parameters, such as learning rate and model architecture, are utilized by the system to enhance accuracy throughout the training process [2]. Moreover, rigorous testing across multiple data segments is conducted to guarantee reliability, thereby preventing sample memorization and ensuring identification of genuine patterns. This methodology facilitates the development of effective instruments aimed at mitigating health risks associated with posture and enhancing predictive capabilities.

2.2.3 TensorFlow-based Posture Prediction with MediaPipe Data

This study explores the creation of posture prediction models with MediaPipe and TensorFlow. The predictive model is trained using a time series of skeletal landmark data (such as joint locations and inter-joint angles) that MediaPipe's pose estimation capabilities give. This raw data goes through a preprocessing step before model training, which includes data cleaning, feature engineering (such as determining pertinent kinematic parameters), and normalization/scaling to maximize model performance. Recurrent neural networks (RNNs), especially Long Short-Term Memory (LSTM) networks, are commonly used due to their inherent capacity to process sequential input and capture temporal relationships within postural dynamics [2].

2.2.4 Postural Deviations Using TensorFlow and MediaPipe Data

With TensorFlow and MediaPipe, this work explores the creation of posture prediction models with an eye on future postural deviations before they cause chronic problems. By means of a time series of skeletal landmark data e.g., joint locations, angles, and velocities mediaPipe's pose estimate features capture the dynamics of an individual's posture across time. These expected deviations enable timely interventions that is, feedback (visual, aural, or tactile) given to the user that causes them to adjust their posture before discomfort or pain results. By spotting and correcting inefficient posture patterns early on and encouraging better long-term postural habits, this proactive method seeks to stop the onset of chronic musculoskeletal conditions. Additional studies investigate optimizing model design, including more data sources (e.g., pressure sensor data, activity levels), and assessing the long-term efficacy of these predictive systems in real-world scenarios [2].

2.2.5 Medical and Ergonomic Benefits within Computer-Centric Work Models

Gaps persist within real-time monitoring and evaluation, investigating these areas and validating crucial areas to automate will assist in advancing robust post-evaluation models or frameworks. Proper leverage of posture related datasets elevates machine learning to assess and predict risk trajectories, removing a staggering reliance towards subjective analysis and focusing more on validating refined ergonomic solutions that can continue to adapt over time [3]. Moreover, improving upon these solutions ensure that health benefits can be met and properly build upon existing methodologies as foundational contributions. Ultimately integrating an iterative approach of strengthening proactive prevention, significantly assisting towards worker healthcare.

2.2.6 Office Wellness Programs for Posture Improvement

The increasing prevalence of sedentary work environments has brought the issue of postural health to the forefront of organizational wellbeing initiatives. Prolonged periods of sitting and computer use contribute significantly to musculoskeletal disorders (MSDs), impacting employee health and productivity. Consequently, organizations are increasingly implementing comprehensive wellness programs designed to address postural concerns. These programs often incorporate a range of strategies, including ergonomic assessments of individual workstations to optimize equipment placement and minimize strain, structured stretching and exercise programs to promote flexibility and alleviate muscle stiffness, educational workshops to enhance posture awareness and understanding of proper body mechanics, and the provision of ergonomic equipment such as adjustable chairs and standing desks. These multifaceted approaches reflect a growing recognition of the importance of proactive intervention in mitigating the negative effects of poor posture [4].

2.2.7 Effects of Lifestyle on Body Posture

Poor body posture may be caused by different things and a persons' lifestyle is one of them. Insufficient physical activity and a sedentary lifestyle is closely related to having poor body posture and the behavioral habits of people just like college students have a significant effect on body posture. Relating this to office workers who are using technology for prolonged hours, it is a concern that should be noticed because using gadgets or electronic devices gradually increases the risks that can occur due to poor body posture and because using these devices mostly promotes a sedentary lifestyle that leads to prolonged hours of technology use that could lead to poor body posture. The risks that may occur because of this could be decreased lung capacity, body degeneration, and more [5].

2.2.8 The Psychological Impact of Poor Posture

Addressing the psychological aspects of poor posture requires a holistic and integrated approach. While ergonomic solutions, like optimized workstations and supportive furniture, are crucial, they are insufficient on their own. Effective interventions should also include behavioral strategies that promote posture awareness and support the formation of healthier postural habits. While awareness campaigns are essential in informing individuals about the relationship between posture and psychological health. By making a greater understanding of this connection, influence on other individuals to take proactive measures to enhance their posture, improving their physical and mental wellbeing. Combining the ergonomic adjustments, behavioral changes, and educational initiatives, this multifaceted strategy is crucial for tackling the complex issues linked to poor posture [7].

2.2.9 Body Posture Correction Using Computer Vision

Prolonged improper body posture may result in different kinds of health issues or musculoskeletal disorders that can be detrimental to our health. The study aims to design a

computer-vision based posture correction system for human pose detection. It is a system that reminds users of incorrect body posture with the use of a camera and voice broadcast technology that is paired with an alarm to notify the user if the posture is bad so the user knows when to adjust their posture. The body posture assessment in the study is done by using an algorithm that preprocesses the position information of human key points into feature values which assess the sitting posture based on the support vector machine (SVM) method [8].

2.2.10 MediaPipe and TensorFlow for Posture Classification in Office Environments

In order to identify trends and connections between past posture data and subsequent postural aberrations, the TensorFlow model is trained. Upon training, the model can receive a sequence of recent posture data points as input and generate a prediction of future posture. These anticipated abnormalities make it easier to apply prompt interventions, such as providing the user with feedback (visual, aural, or tactile), which promotes postural corrections and delays the beginning of pain or discomfort. The approach's main goal is to prevent chronic musculoskeletal problems by proactively detecting and correcting inefficient posture patterns early on. With that, it will encourage the formation of healthier long-term postural habits. The methodology's approach, ability to provide feedback, capabilities, and potential for long-lasting posture improvements are some of its possible advantages [9].

2.2.11 TensorFlow-Based Neural Networks for Posture Classification

The advent of TensorFlow has revolutionized the field of automated posture classification, enabling the development of sophisticated deep-learning models capable of accurately identifying various human postures. Researchers have leveraged TensorFlow's capabilities to train both deep neural networks (DNNs) and convolutional neural networks (CNNs) on extensive datasets encompassing a wide range of postures, from proper sitting and

standing to more problematic positions like slouching and forward head posture. This allows for a nuanced understanding of postural variations and deviations from ideal alignment. Furthermore, applying transfer learning techniques and utilizing pre-trained models such as MobileNet and EfficientNet have proven highly effective. This approach enhances classification accuracy and significantly reduces the computational resources and training time required, making these systems more practical and accessible [9].

2.2.12 Machine Learning through Computer Vision

Computer vision extracts information from images or videos which acts like human vision. Machine learning on the other hand can train a dataset to learn from a dataset which can be used to extract useful information through different testing. One of the major research areas of computer vision is human pose estimation which analyzes the pose of humans. The research problem in this area of computer vision is to accurately identify the right location of human joint or anatomical keypoints. Human poses can also vary which makes it hard to estimate poses. Machine learning could aid in resolving this problem by training multiple datasets that show different kinds of human poses and learn to accurately identify poses that humans do. The study shows the importance of machine learning in image processing and computer vision, where machine learning is spreading to multiple research areas of computer vision and is utilized well and other research areas or fields are open for machine learning to be used [10].

2.2.13 TensorFlow and Its Application in Posture Assessment

TensorFlow, an open-source machine learning framework developed by Google, has revolutionized large-scale machine learning applications, including computer vision and deep learning-based posture assessment. The comprehensive libraries and adaptable architecture of TensorFlow enable the development and implementation of complex neural network models that

can estimate human pose in real time. TensorFlow's features can be used to create intelligent posture analysis and correction systems, which is important given the growing worries about extended occupational sitting and its effects on health [10].

2.2.14 Future Directions in Machine Learning for Posture Analysis

The world of posture analysis is about to get much smarter, thanks to some exciting advancements in machine learning. Imagine a future where technology can seamlessly monitor your posture throughout the day, offering personalized feedback and helping you avoid those nagging aches and pains. That is where things are headed, advancements like federated learning, which allows AI models to learn from many users without compromising privacy, and edge computing, which brings the processing power closer to the device, making posture analysis faster and more efficient. Tools like TensorFlow, YOLOv5, and MediaPipe are constantly being updated, giving us even better ways to recognize and understand posture [12].

2.2.15 Combination of MediaPipe and YOLOv5

MediaPipe is a framework that allows access to advanced pose estimation and motion analysis capabilities that allows us to detect and track body posture. It is an open-source machine learning framework which is primarily used for human pose estimation. MediaPipe is a powerful tool for posture detection and with the aid of computer vision, it is able to accurately identify proper body posture. YOLOv5 on the other hand is an algorithm that efficiently detects objects and posture. It is able to detect these objects fast and accurately in different sizes and types. Combining both MediaPipe and YOLOv5 will lead to much more detailed and comprehensive results which are more accurate, leading to better analysis of poor or good body posture [12].

2.2.16 YOLOv5 and MediaPipe for Enhanced Pose Estimation

In order to overcome the drawbacks of utilizing each technology alone, this study investigates the synergistic combination of YOLOv5 and MediaPipe for improved posture estimation. To locate and identify the entire body region in an image or video frame, YOLOv5, which is well-known for its precision in object detection, will be used. As a vital pre-processing step, bounding box identification efficiently reduces the search space for ensuing posture estimation. Within the area that YOLOv5 discovered, MediaPipe which specializes in sophisticated bone landmark detection is then used. By using this focused strategy, MediaPipe may concentrate its processing resources on a more manageable and pertinent region, increasing accuracy and efficiency. The combination seeks to improve pose estimation's resilience, especially under difficult situations [12].

2.2.17 The Impact of Stress on Posture

Whether seen as an emotional burden or a more general effect on worker wellbeing, occupational stress has a major impact on posture. Many office-based employees report medium to high degrees of stress, which might show up as different postural abnormalities. These stress-related effects could be slouching, a flexed or bent posture, or conversely, an unduly extended or "open" posture, both possibly aggravating musculoskeletal pain and long-term health problems. Stress and posture have a complicated, multifarious relationship. Muscle tension brought on by stress influences postural control and alignment by itself. People under stress, for instance, could unintentionally stiffen their shoulders, resulting in rounded shoulders and forward head position [14].

2.2.18 The Role of Posture in Physical Health

Good posture is important for preserving a healthy skeletal system in your body. As stated in a study by Rush University, correct posture helps to maintain the spine's natural "S"

curve, minimizing the strain on muscles and ligaments around the areas of the body, which aids in preventing injuries and chronic discomfort. The same as from, Harvard Health it highlights that tight muscles in the hips and chest can interfere with posture, resulting in pain and reduced mobility. Strengthening the core muscles is vital for correcting posture. Exercises that focus on the transverse abdominis, obliques, and glutes can enhance spinal alignment and lower the likelihood of experiencing lower back pain. Adding routines such as planks, glute bridges, and bird-dog exercises can mitigate the negative effects of extended periods of sitting [16].

2.2.19 The Impact of Work Lifestyle on Posture

Research emphasizes the importance of several key strategies to mitigate these risks. Firstly, establishing an ergonomic home workspace is crucial, including investing in adjustable chairs, monitors positioned at eye level, and keyboards/mice that promote neutral wrist posture. Secondly, incorporating regular movement breaks into the workday is essential. This can involve setting reminders to stand up and stretch, engaging in short walks, or participating in online exercise classes. Finally, the use of sit-stand desks, which allow for alternating between sitting and standing throughout the day, has been shown to be effective in counteracting the adverse effects of prolonged sitting and promoting better posture. Further research is exploring the long-term impacts of remote work on posture and the effectiveness of various interventions aimed at promoting ergonomic awareness and healthy postural habits in the work-from-home context. The challenge lies in empowering remote workers with the knowledge and resources to create healthy work environments within their homes [21].

2.2.20 Posture and Mental Wellbeing

Beyond the physical benefits of good posture, it positively influences good mental health. Research published in Health Psychology shows that individuals maintaining upright posture

experience enhanced mood, increased self-esteem, and reduced stress levels . WebMD also notes that proper posture can boost energy levels and self-confidence, contributing to overall wellbeing. Additionally, maintaining an upright posture can alleviate feelings of fear and anxiety, promoting a more positive outlook and emotional resilience [22].

2.2.21 The Impact of Work-From-Home Arrangements on Posture

The rapid shift towards remote work arrangements has brought heightened concerns regarding posture-related health issues. A significant contributing factor is the prevalence of non-ergonomic home office setups, often lacking the adjustable furniture, proper lighting, and optimized equipment placement found in traditional office environments. This deficiency in ergonomic design, coupled with the tendency towards prolonged sitting due to reduced commuting time and the blurring of work-life boundaries, has demonstrably contributed to a rise in musculoskeletal complaints among remote workers. Studies have shown a correlation between poorly designed home workstations and increased reports of neck pain, back pain, and other musculoskeletal discomfort. The lack of movement between breaks, often facilitated by the office environment (e.g., walking to meetings, interacting with colleagues), can lead to negative effects of prolonged sitting [24].

2.2.22 Posture Recognition Systems in Office Environments

Office environments present some unique challenges to posture recognition, since systems for these environments need to be nonintrusive, efficient, and operate continuously. Many publications depict how posture monitoring can be integrated into routine office activities, such as "Automated Detection of Improper Sitting Postures in Computer Users Based on Motion Capture Sensors.". These systems often utilize motion capture or pressure sensors, not disturbing workflow, so that the employees can receive posture corrections seamlessly. Office-centric

systems can thereby enhance employee wellbeing and reduce the prevalence of musculoskeletal disorders from seating-related activities [25].

2.2.23 The Impact of Prolonged Occupational Sitting on Musculoskeletal Health

Prolonged sitting, categorized as sedentary behavior, has been extensively studied for its adverse health effects, particularly for musculoskeletal complaints (MSCs). Sedentary behavior, including extended sitting hours in occupational settings, has been linked to various health risks, including chronic pain and discomfort in the neck, shoulders, upper back, and lower back (S-MGA Study, 2022). The study conducted as part of the Study of Mental Health in the Workplace (S-MGA) in Germany examined the prevalence of MSCs associated with prolonged sitting [27].

2.2.24 Defining Prolonged Occupational Sitting and Its Health Implications

Prolonged occupational sitting is increasingly recognized as a significant health concern due to its association with various physiological and cognitive impairments. Sedentary behavior, which includes extended periods of sitting, has been linked to adverse health outcomes such as cardiovascular disease, metabolic disorders, and musculoskeletal complaints. Studies indicate that desk-based office workers spend approximately 70–85% of their work hours sitting, contributing significantly to their daily sedentary time.

Prolonged sitting will be defined as sitting for more than four hours without significant movement, with durations exceeding six to eight hours per day will lead to severe health risks. Such excessive sitting times has been associated with impairments in executive function, memory, and attention, which are crucial for work productivity [27].

2.2.25 Use of Machine Learning for Musculoskeletal Disorder Prevention

Analyzing posture is one of the possible ways to prevent musculoskeletal disorders. That's why studies recognized the importance of posture recognition systems and adapted different systems such as MediaPipe to recognize different poses and collected data to help with posture analysis. The data is collected from various people of different ages, sex, height, and body class to receive a more accurate representation. The data collected confirmed that the use of different posture recognition systems can help in assessing healthy and proper body posture and help health professionals in treatment in terms of physical therapy and ergonomics.

How people engage in regards to prevention of muscular or joint problems are further analyzed by neural networks, gathering and processing inputs from wearable sensors or camera. Assessing features of spinal curvature and weight distribution, focusing on subtle deviations, such as asymmetrical shoulder alignment or excessive lumbar flexion. Integration of predictive analytics counteract habitual and harmful postures, alerting users while also utilizing adaptive ergonomic insights. Research indicates that machine learning tools can benefit high-risk groups such as office works, reducing muscle and joint issues due to clinicians leveraging analysis systems to design personalized ergonomic health plans that accompany individual needs. In a study of 200 participants, utilizing MediaPipe Pose with a BlazePose and MobileNetv2 architecture. Repeated trials detected 33 anatomical landmarks, mapping key points throughout shoulders and spine, proving high reliability and consistent results by finding clear links between factors such as height, shoulders, or spinal misalignment. Although beneficial in its preventive and clinical nature, these tools need to address financial cost and ease of use for broader adoption [28].

2.2.26 MediaPipe's Role in Skeletal Tracking for Posture Monitoring

MediaPipe, an open-source framework developed by Google, has gained recognition for its efficient skeletal tracking capabilities, which are crucial for posture monitoring applications.

MediaPipe's Pose module can detect key skeletal landmarks using machine learning, enabling real-time posture assessments through mobile and web-based applications. Studies have shown that MediaPipe-based posture monitoring systems can provide instant feedback to users, encouraging ergonomic sitting and standing habits. The lightweight nature of MediaPipe allows for seamless integration into consumer wearables and workplace health initiatives [28].

2.2.27 Impacts of Workplace Posture on Ergonomics and Blood Circulation

Poor circulation from stagnant workplace environments negatively impacts both ergonomic comfort and circulation stability. Understanding circulatory strain as a significant factor in overall discomfort and wellbeing leads to conversations in which posture shifts can improve blood flow. Research indicates that integrating movement and varying postures in the workplace assists in healthy blood circulation, mitigating reduced circulation. Moreover, regular adjustments in posture throughout the workday are advantageous as they mitigate the risks of neck fatigue and blood pooling in the lower limbs [29].

2.2.28 Health Implications of Poor Sitting Posture

Poor body posture is said to contribute to numerous health conditions, the common being musculoskeletal aches, exhaustion, and loss of efficiency. Research has also further indicated that improper postures cause harm to one's health in terms of physical complications, among which are symptoms of back pain, neck tension, and headache. For instance, a study on "Perceived Musculoskeletal Discomfort and Its Association with Postural Shifts During Prolonged Sitting in Office Workers" found a significant relationship between postures and chronic pain conditions. According to the research findings, there is a need for systems capable of identifying adverse postures that are corrected to forestall such ill health effects, especially among office workers, who are at a greater risk of sitting for extended periods of time [30].

Prolonged static postures in office environments can severely disrupt spinal alignment and strain the joints, such imbalance creates wear-and-tear throughout the body, at points where movement worsens or gets limited. Workers with continued slouching or unnatural postural positions during work sessions face greater risk of accumulating spinal damage, workspace environments that poorly adjust to the ergonomic need of workers compound upon these risks. Consideration towards posture monitoring and ergonomic training initiatives combat and prevent these potential risks from becoming chronic.

2.2.29 Reinforcing Optimal Posture Perceptions

Physical therapists have a subjective definition towards the judgement of posture, agreeing on the ideal posture without precise measurement but rather from experience as a trained professional. To understand more accurately what differentiates an optimal posture from a harmful one, a newer basis for precise measurement must be established to eliminate visual bias among professionals. There are current limitations from assessing posture through a static viewpoint, and a solution that provides dynamic feedback for a variety of posture shifts enhances objectivity within movement analysis. Although challenging their beliefs, presenting the necessary measurement tools can shape and level how practitioners perceive optimal posture [31].

2.2.30 Multifactorial Approach to Habitual Sitting Posture and Movement Variability

Prolonged sitting leads to unavoidable changes in posture because the body naturally seeks to alleviate localized strain, often settling into positions that prioritize comfort over optimal posture. Studies indicate that it is not feasible to maintain a single static posture continuously; rather, individuals make micro-adjustments to prevent discomfort [33]. Individuals often find themselves in flexed sitting postures after long work sessions, making slight adjustments to more

rigid positions in an attempt to alleviate back pain. Underscoring the nuance in movement variety can assist in maintaining a dynamic approach to sitting, preventing awkward positions that overwork muscle groups.

2.2.31 Assessing Smart Sensor Effectiveness on Posture Correction

When it comes to employment, there are many different scenarios in which sensor systems can be utilized effectively for those who suffer from musculoskeletal disorders (MSDs). In these contexts, a prioritized feedback system is necessary. According to the findings of the study, in order to get a high classification accuracy rate for the purpose of accomplishing successful targeted changes, it is necessary to use a variety of machine learning models when evaluating multiple backgrounds, such as health, demographics, or body form [34].

2.3. Gaps in Existing Literature

While posture recognition systems have demonstrated feasibility, significant gaps remain, especially in real-time corrective feedback and accessibility. Many current solutions are not adaptable to localized office environments, such as those in the Philippines, where ergonomic interventions are often limited. There is also a need for continuous, unobtrusive feedback that considers cultural and spatial constraints in office settings. This study aims to address these issues by developing an accessible, real-time posture recognition system that can be easily integrated into offices and provide users with prompt and post-corrective feedback.

2.4. Summary

Studies conducted on machine learning-driven posture recognition offer important methodologies, especially through the application of visual models such as MediaPipe and proficient machine learning methods that improve precision and effectiveness in identifying postural deviations. The studies emphasize both the advantages and drawbacks of existing pose estimation methods, showcasing their ability to enhance workplace ergonomics while also pointing out their vulnerability to factors like environment and human intervention.

The review examines the effectiveness of machine learning and algorithmic technologies for posture recognition and human activity detection. While progress has been made in such through the findings and literature that have been compiled, gaps remain in the practical application and tailoring of posture monitoring and feedback systems, particularly within office environments and in a post feedback based feature. Current literature highlights challenges in providing feedback and adapting these systems to the specific conditions of office work. The research aims to address these limitations by developing a more practical posture post-recognition algorithm that mitigates health risks associated with prolonged sitting and poor posture among office workers which will better-mine the overall office environment and the human lifestyle.

Chapter 3

3.1. Conceptual Framework

The procedure described in this chapter outlines a structured approach to developing and evaluating an innovative posture recognition algorithm, responding to growing concerns about the health effects of prolonged poor posture in office settings. To address these concerns, the algorithm must deliver immediate feedback to users on posture correction using MediaPipe, Tensorflow, and YOLOv5. Data collection, pre-processing, model development, and algorithm

assessment are discussed in detail to ensure rigor and reproducibility. Using the accuracy and responsiveness of MediaPipe and Tensorflow helps ensure correct input, which may gradually reduce the risk of musculoskeletal problems from poor posture. This approach allows for assessing the algorithm's effectiveness in improving posture awareness, identifying monitoring challenges, and evaluating progress in posture habits.

Ensuring that the algorithm is trained on both correct and incorrect sitting positions, the datasets should cover a wide variety of sitting postures with the help of a professional to assist in identifying proper and improper postures. The posture landmarks that will be considered are mostly upper body landmarks that center around the spine as this is sufficient in detecting good or bad sitting postures. Only necessary features will be considered as “Hughes Phenomenon” could happen where it means that adding beyond the optimal number of features will possibly degrade the model’s performance. If publicly available datasets are insufficient, data will be systematically collected from office or workplace environments to accurately capture authentic office configurations, including desk arrangements, lighting conditions, and seating layouts that may affect posture. The selection of data sources will be guided by data quality, diversity of postures, and relevance to office settings. Public datasets should include images captured in work contexts, while custom data collection will prioritize authentic office poses and representation of diverse demographics.

Figure 3 illustrates the conceptual framework for a posture assessment system integrating YOLOv5, Tensorflow and MediaPipe. The process begins with gathering data, which will come from datasets available online which will be used for model training and test dataset that is gathered by the researchers in an office environment to test the effectiveness of the model. The prepared data will then undergo an extraction process where MediaPipe identifies the key points and anatomical landmarks from the prepared dataset. After extraction, the model that will be used will undergo training to enhance its capabilities and to teach it how to identify good and bad

posture from the dataset provided using Tensorflow. The trained model will then be used to identify good and bad posture and give corrective feedback alongside YOLOv5 that is used for only detecting a person. If the posture is identified as sitting, the system further classifies it as good or bad posture. If the posture is good, MediaPipe still processes the pose, but the system verifies the proper posture. The framework operates continuously, analyzing each data frame to ensure post posture monitoring and assessment and will only end if the worker stops working. The testing of the model will be done by providing a recorded video of an office work environment where it can be used to know if the model is working effectively. The model should provide a proper corrective feedback and will continuously run until the workers stopped working or if it is stopped.

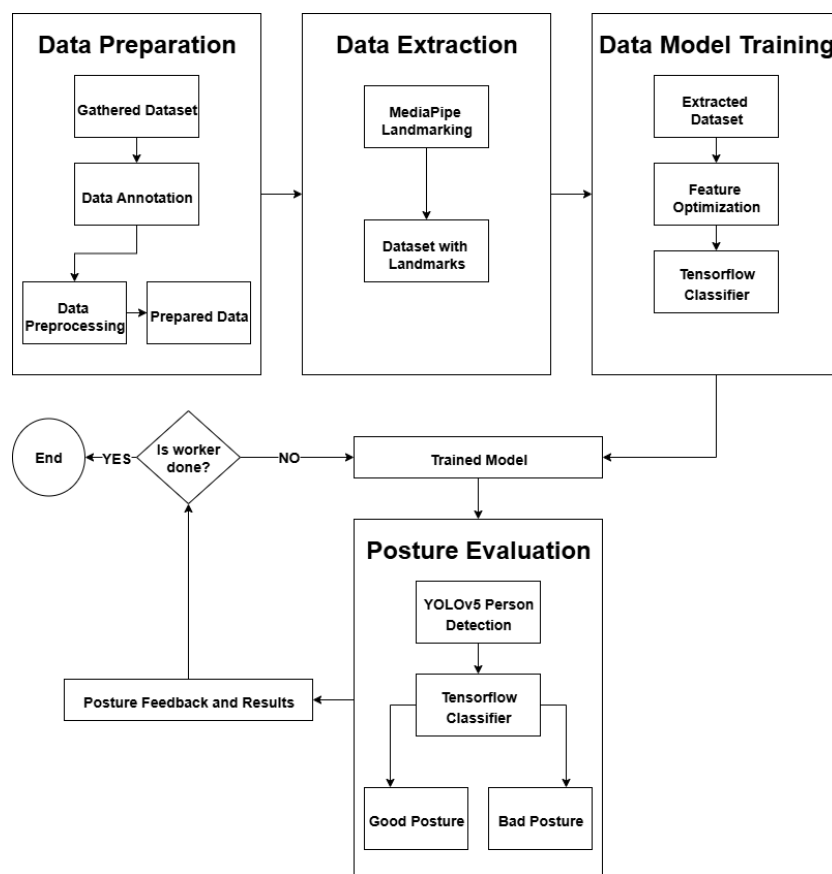


Figure 3. Conceptual Framework

3.2. Data Gathering

Data for this study will primarily be gathered from publicly available datasets through platforms like Roboflow universe and additional online resources, ensuring a diverse and thorough collection of posture images but some data will also be gathered to local office environments to be used for testing the model. These datasets will feature labeled images showing individuals in various sitting postures, categorized as correct or incorrect, to efficiently train the machine learning model with the assistance of a health professional to properly annotate what posture is correct or incorrect. The choice of datasets will emphasize high-resolution images captured in real-world office environments to improve the system's capability to identify posture variations accurately. In contrast to Kaggle, which is frequently utilized for structured datasets, Roboflow Universe offers specialized image datasets that are more closely aligned with the requirements of this study's deep learning methodology.

Additional image information may be gathered from online platforms that offer open-access posture datasets to enhance the available datasets, guaranteeing a broader assortment of postural differences. This action is crucial to tackle possible deficiencies in current datasets and enhance the model's flexibility to various office settings. The collected images will be pre-processed, annotated, and augmented to enhance model robustness, ensuring the system can effectively identify good and poor sitting postures across diverse scenarios. Various lighting conditions, desk arrangements, and seating layouts will be considered to improve real-world applicability.

Developing an effective posture recognition system necessitates guaranteeing the diversity and quality of the data. The integration of TensorFlow and MediaPipe will facilitate posture identification using the gathered datasets. By employing AI-driven methodologies, the model will perpetually learn from real-world instances, enhancing its ability to provide timely

remedial feedback. This systematic data gathering and processing approach will help create a more accurate, responsive, and accessible posture monitoring system tailored for office environments.

3.3 Data Pre-Processing

Preparing and implementing raw posture data requires substantial analysis and effective training to create a reliable and accurate dataset for office postures. Handling this dataset needs essential techniques like data cleaning and transformation. These steps help build a robust dataset with accurate and diverse characteristics for a relevant overview of office postures.

To ensure data integrity and relevance, several data cleaning procedures will be implemented. Initially, frames or images that do not clearly depict a person's posture, such as those without a seated individual or with obscured postures, will be excluded to eliminate noise and irrelevant samples that may hinder model performance. For accurate posture identification, the model will be trained on clearly defined postures by removing or annotating frames in which anatomical features are not visible. Duplicate frames will also be eliminated to reduce redundancy, prevent overfitting, and ensure balanced representation of various postures, particularly in dynamic posture video recordings.

3.4. Data Split

The dataset that is sourced publicly will be split into 70%-20%-10% format where training will take up 70%, validation will take 20%, and testing will take 10% of the split. The data split choice is sufficient as it is a widely accepted value and it ensures that the model is

trained with a large volume of data while also having sufficient checks to the model's performance while also having an overall evaluation of the model to unseen data.

3.5. Algorithm Training

The algorithm will be trained with MediaPipe and Tensorflow to identify body posture accurately. The YOLOv5 pre-trained model is used for person recognition, and the TensorFlow-trained classifier will be the basis for the posture assessment system. The model will utilize publicly accessible datasets from sources like Roboflow universe and other online platforms, ensuring a wide variety of around 500-1000 labeled images that depict proper and improper sitting postures for training.

Before training the data, the dataset used should be carefully preprocessed to enhance its quality and eligibility for the model. The data cleaning techniques will remove any unnecessary, irrelevant, or duplicate images affecting the model's accuracy. The data augmentation methods like rotation, scaling, and contrast changes will be implemented to create variations in postural representations, thus making the model more robust against various office conditions. The dataset will also be improved to include variations in lighting situations, desk setups, and seating positions, ensuring the system can effectively identify postural deviations.

Training the algorithm will utilize TensorFlow's deep learning capabilities and MediaPipe's landmark extraction. The model will undergo iterative training using supervised learning, where labeled posture images will guide its classification process. Supervised learning is used because the model can learn efficiently what is improper and proper posture easily from the labeled dataset provided, which can be found at Roboflow universe. Unsupervised learning is

not used because the model will have a much harder time learning, as it does not have a proper identification, since it only has unlabeled data, which does not help the model learn efficiently. The training process will consist of optimizing hyperparameters, modifying learning rates with ADAM(Adaptive Moment Estimation), and utilizing loss functions like binary cross entropy to improve the model's ability to distinguish between good and bad posture. After the model attains a satisfactory level of accuracy, it will be tested with unseen data to verify its reliability in real office situations. Using continuous learning and updates, the trained algorithm will boost its effectiveness in providing corrective feedback, promoting better posture practices in workplace settings.

Figure 4 shows two branches of machine learning, which are Supervised learning and Unsupervised learning. It shows one key difference, which is the use of labeled and unlabeled datasets, in which Supervised learning is used in the study because it uses a labeled dataset, which can help the model be trained more efficiently.

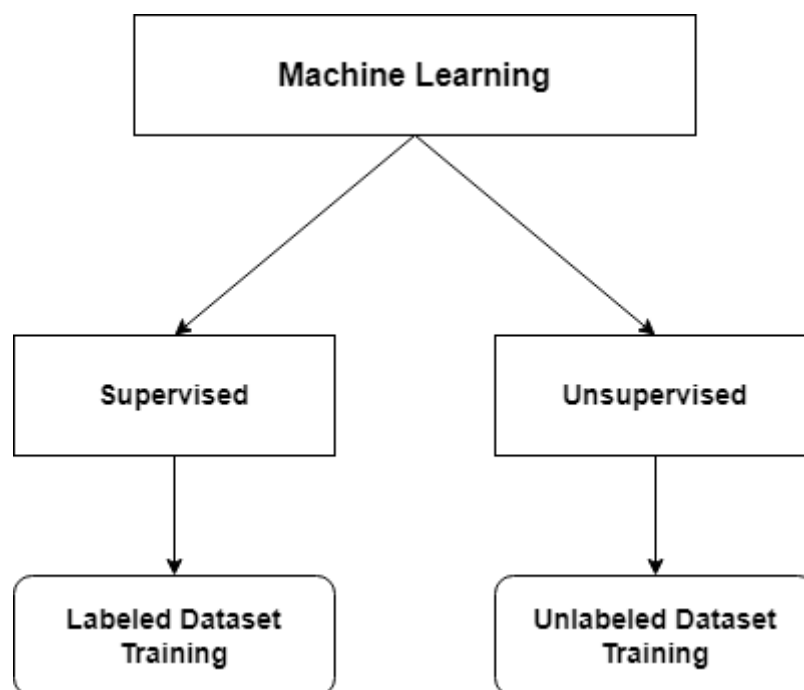


Figure 4. Machine Learning

3.6. Implementation and Data Preparation

3.6.1 Data sources and collection

The data set combines labeled images from public repositories (e.g., Roboflow Universe) and recordings from actual office environments. To ensure the model learns both proper and improper sitting postures, a health professional verified the class labels. This emphasizes expanding the data set with video recordings of team members and colleagues, especially in challenging scenarios (chairs with armrests, varied lighting) identified during preliminary tests. All data gathered from the researchers' own sources will be collected only from participants who have provided informed consent through a formal consent form. The researchers will ensure that all personal and visual data are handled in strict compliance with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173). Collected data will be stored securely, used solely for research purposes, and properly disposed of after the study through systematic deletion and data cleaning procedures to protect participants' privacy and confidentiality.

3.6.2 Data Assurance

Professional Physical Therapists from both public and private clinics validated the collected data to make sure it was accurate. These experts gave their professional opinion to confirm that the posture classifications used in the dataset were correct and the wrong ways to sit. The experts' involvement helped make sure that the training data was very accurate, which made the model's recognition process more reliable and in line with real-world ergonomic and physiological principles.

3.6.3 Data Cleaning and Augmentation

The cleaned data set is created by removing frames without visible subjects or with occluded landmarks. Duplicate frames are eliminated to reduce bias. Augmentation procedures – rotation, flipping, scaling, and brightness adjustments – expand the training data to simulate real-world variation and mitigate the relatively small sample size (500–1,000 images). All augmented images maintain the original file names to stay synchronized with their label files.

3.6.4 Model architecture

The core of the posture recognition system remains the integration of YOLOv5 for person detection, MediaPipe Pose for landmark estimation, and Tensorflow for posture classification. YOLOv5 isolates the seated subject in the frame, and MediaPipe returns the 3D coordinates of 29 body landmarks to help Tensorflow determine the good and bad postures. From these coordinates, angles and ratios between key points (e.g., head–shoulder–hip angle) are calculated and fed into a lightweight neural network classifier. This classifier outputs a binary prediction (good or bad posture) and forms the basis for corrective feedback.

Landmark Name	Feature Set
Nose	X Nose, Y Nose, Z Nose, V Nose
Shoulder	X Shoulder, Y Shoulder, Z Shoulder, V Shoulder
Elbow	X Elbow, Y Elbow, Z Elbow, V Elbow
Wrist	X Wrist, Y Wrist, Z Wrist, V Wrist
Hip	X Hip, Y Hip, Z Hip, V Hip
Knee	X Knee, Y Knee, Z Knee, V Knee
Ankle	X Ankle, Y Ankle, Z Ankle, V Ankle

Figure 5. Features

The 29th feature is the torso tilt angle, which is the measure of the vertical line connecting the mid-shoulder and mid-hip centers, indicating if a person is leaning too far forward or backward.

X values stand for the horizontal location of the joints.

Y values stand for the vertical location of the joints.

Z values stand for the distance of the joint from the camera.

V values stand for the confidence score of the visibility of the landmark.

The values of the features will play a critical role in the conversion of 3D human pose to a format that neural networks can understand and analyze. It will determine the good and bad postures from the data given, which will help achieve the goal of the study.

3.6.6 Training procedure

The classifier will be trained using supervised learning with a cross-entropy loss function. Hyperparameters such as learning rate, batch size, and network depth will be tuned on the validation set. Early stopping and regularization will be used to prevent over-fitting. Because the data set is still being prepared, this subsection currently outlines the planned training configuration; training curves and quantitative results will be inserted once training is complete.

3.7. Detection Results and Examples

This section provides qualitative observations of the developed posture recognition model after close testing, iteration, and successful training. The findings highlighted the model's ability to classify or accurately demonstrate practical performance within office environments. Visualizations, gaps in classification, and further examples of corrective feedback are discussed comprehensively to fully understand the system's real world capabilities.

3.7.1 Sample of Good and Bad Posture Detection



Figure 6. Posture Classifications

Figure 6 shows the detection of good or bad posture in an office setting, highlighting the key points of the shoulder and the arrow indicating the spine. The figure also actively records the amount of good or bad detected posture to give a user an idea as to when they should adjust their posture.

3.7.2 Detection Mismatch

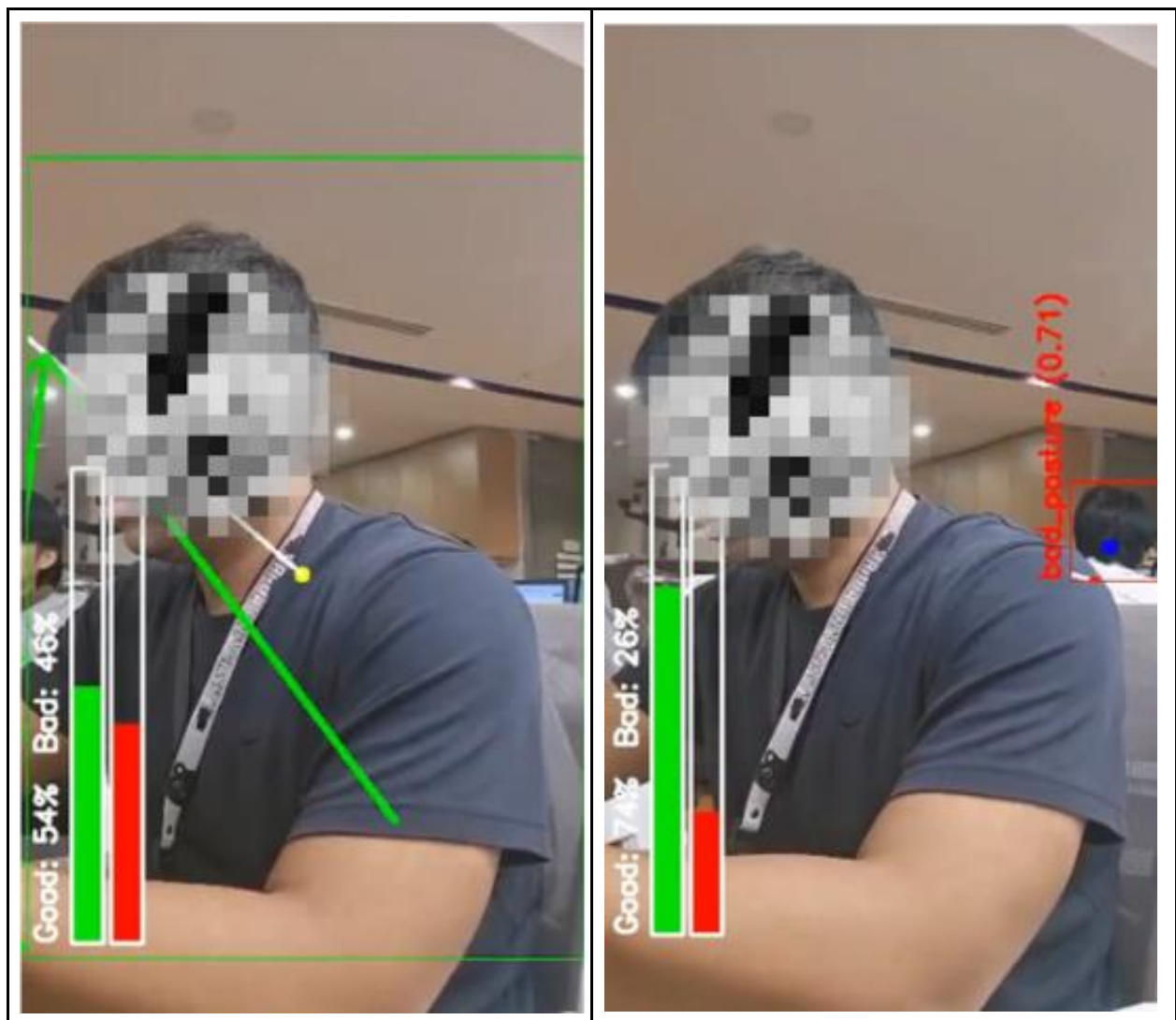


Figure 7. Detection Error

Figure 7 shows frames where detection is off or not of the intended person. This shows that the model could still sometimes have imperfections where it could not recognize the person in the frame, or it could also give some false detections within certain conditions, like an obstructed view.

3.7.3 Post Corrective Feedback



Figure 8. Feedback

Figure 8 shows the post-analysis and feedback of the model, where it appends seconds of frames at the end of the inferred video to show the analysis of the whole video, and provides feedback on whether the user is doing well in keeping a good posture or not.

3.8. Comparative Analysis

Examining how well the optimized model performs in comparison to its internal configurations and similar studies on upper-body position recognition is the main objective of the comparative analysis. Internally, the researchers will assess the fully integrated optimal

model, a configuration that simply extracts landmarks, and a detection-only configuration to see how they all affect the system's behavior as a whole. Despite its effectiveness in recognizing joint positions, the landmark-extraction setup is more likely to have localization mistakes and inconsistencies in surroundings that are more complex. The improvements in the model's streamlined design for detection, landmark estimation, and classification contributes to its stability, accuracy, and real-time performance. The analysis that follows will use metrics like AUC, loss, and accuracy to put a number on these enhancements.

The optimized model will undergo internal review as well as comparisons with previous research on upper-body posture classification. Many other strategies have been proposed in recent research, including deep learning systems trained on images and hybrid approaches that use both visual and sensor-based inputs. To evaluate the performance of the optimized model, these studies give reference values, usually expressed as accuracy, precision, recall, or mean average precision. Methodological variations, such as depending on standard video data and using only upper-body landmarks, impact generalizability and robustness, and this will be highlighted in the comparison.

By taking this approach, the researchers gain more knowledge about how the improved model advances non-invasive, camera-based ergonomic monitoring and how it aligns with current research priorities. The analysis will use this framework to determine how much optimization improves classification performance and whether the suggested method is viable in the upper-body posture identification field overall.

3.9. Interpretation of Results

The improved posture recognition model was predicted to show great flexibility and precision in actual office situations before getting the last results. The model aimed to address changes in lighting conditions, chair arrangements, and user behavior commonly observed in BPO centers and IT offices, among other settings. The system's architecture is designed to focus on upper-body markers, which are the following: head, neck, shoulders, and spine, allowing it to accurately detect poor posture without requiring any specialized equipment. This arrangement was designed to enable the model to be useful for implementation with typical office webcams, thereby offering a non-invasive approach to promote ergonomic awareness and posture correction among staff.

Encouraging correct sitting posture and reducing the risk of musculoskeletal disorders associated with prolonged improper posture should help improve office ergonomics. Giving customers the correct comments helps them to make modest, daily tweaks that will gradually improve posture consistency and general workplace wellbeing. The data-driven approach of the model was also expected to assist businesses in spotting posture patterns and creating ergonomic solutions that would boost long-term comfort and productivity

Before testing, though, several restrictions were projected. Using YOLOv5 with MediaPipe may cause computer lag, which could slow down real-time recognition, particularly on devices with little processing capability. The model was also designed to be sensitive to occlusion, such as when objects, tables, or armrests partially obscure users' upper bodies, which may reduce the precision of crucial point identification. Furthermore, the training dataset mostly concentrated on typical office postures, the model might first have difficulty generalizing across users with different body types, chair designs, or unusual seating positions. These components,

which were recognized as possible challenges, will have to be optimized and need more extensive data collecting in later stages.

3.10. Summary of Findings

The researchers covered every step of the process for creating an ergonomic monitoring system for the workplace that uses computer vision for posture identification. This study integrated YOLOv5, MediaPipe Pose, and TensorFlow into an integrated pipeline, labelled images, and prepared datasets. It also included data-augmentation procedures. The research laid the groundwork for the real-time identification and categorization of upper-body sitting positions utilizing standard RGB video input by employing this methodology.

Preliminary tests indicate that combining detection and pose-estimation networks leads to more consistent and understandable results than using either method alone. The system's modular design allows multiple users to receive adaptive feedback, which supports faster experimentation and future growth. Detailed quantitative results, including accuracy, loss, and comparative performance, will be discussed in later chapters, where the focus will shift to how the optimized model meets research goals of reliability, ergonomic relevance, and accuracy. This chapter, in contrast, has focused on system design and implementation as a basis for subsequent analysis.

3.11. Ethical Considerations

The study obtained datasets from public sources but also from private sources and it has considered the protection of vital information of participants to follow Republic Act No. 10173 or Data Privacy Act of 2012 from the Philippines. In which, datasets obtained from private sources will be blurred out but consent is given in the demonstration of videos only when necessary. All privately obtained data will be deleted at the end or publication of the study.

Chapter 4

4.1 Model Evaluation and Formulas

Results are discussed in this portion and some formulas or visualizations are provided for more clarity. The results that will be presented are from the model that aims to have an accurate reading in determining good or bad posture while sitting in an office environment. For more clarification, some formulas are introduced to show that the results given by the model are satisfactory.

Figure 9 shows the F1-Score formula which is responsible for the calculation of the harmony of precision and recall which are both responsible for the evaluation of the performance of the model. Precision tells us how right or exact the model is in its predictions while recall on the other hand tells us how complete or thorough the model is in identifying actual cases of the classes. F1-Score balances out both precision and recall and gives a score that will show if the model is performing well or not by having a score out of 1 which means “great performance” while having a score of 0 which means “worst performance”.

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Figure 9. F1-Score Formula

Figure 10 is the formula used by the scaler function used in the model that handles normalization or standardization. It is crucial to have this because it ensures consistency between the data that was used for training and the video that is used for testing. It prevents performance degradation and makes the model accurate in its predictions.

“ X ” is the value of one of the 29 features or landmarks used before scaling.

“ μ ” is the average value of the feature or landmark that is calculated from the dataset.

“ σ ” measures the consistency or variability of the feature or landmark across the entire dataset.

“ X_{scaled} ” is the completely processed input fed to the model for equalizing features to prevent other features or landmarks from overpowering other features.

The standard scaler plays an important role in making the model accurate and consistent by enabling fair judgements of landmarks or features while also speeding up and stabilizing model training by acting as a map.

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma}$$

Figure 10. Scaler Formula

4.2 Analysis of Results

This chapter examines the outcomes of the designed posture recognition model's testing and evaluation. It offers a comparative result of the unoptimized, optimized models as well as models from other studies, emphasizing important performance indicators including accuracy, precision, recall, F1-score, and AUC. The study seeks to evaluate the dependability and correctness of the suggested system in correctly identifying good and bad positions as well as to emphasize gains attained by means of optimization. Additionally, to further ensure the model's reliability and practical applicability, the system's performance outputs were reviewed and validated once more by professional Physical Therapists from both public and private clinics, the exact same ones who viewed the training data. Their professional insight helped confirm that the

model's classifications aligned with ergonomically sound posture standards, ensuring that the results reflected not only computational accuracy but also real-world physiological correctness.

4.2.1 Unoptimized Model

Detailed Classification Report (Unoptimized Model)	
Model Accuracy	76.69% (0.7669)
Model Loss	50.62 % (0.5062)
Model AUC	83.61% (0.8361)

Figure 11. Unoptimized Metrics

Figure 11 shows the performance metrics of the Unoptimized Model, which acts as the baseline for measuring improvements in later versions. The model achieved an accuracy of 76.69% (0.7669), meaning it correctly identified posture classifications most of the time, but still made mistakes due to limited optimization. Its loss value of 50.62% (0.5062) indicates that the model had trouble with consistent learning and struggled to reduce prediction errors, likely due to inadequate parameter tuning and data generalization. At the same time, the model's AUC of 83.61% (0.8361) shows a decent ability to distinguish between good and bad posture classes, though it lacks the clear boundaries seen in the optimized version. Overall, these results show that while the unoptimized model performs fairly well, it still needs improvements in training, data augmentation, and hyperparameter tuning to reach higher accuracy, lower loss, and more reliable classification outcomes.

Detailed Classification Report (Unoptimized Model)				
	Precision	Recall	F1-Score	Count
Bad Posture	78% (0.78)	76% (0.76)	77% (0.77)	68
Good Posture	80% (0.80)	77% (0.77)	76% (0.76)	65
Accuracy			77% (0.77)	133
Macro Avg	77% (0.77)	77% (0.77)	77% (0.77)	133
Weighted Avg	77% (0.77)	77% (0.77)	77% (0.77)	133

Figure 12. Unoptimized Class Results

Figure 12 shows the classification results of the Unoptimized Model. It highlights how well the model identifies both good and bad posture classes. For bad posture, the model achieved a precision of 78% (0.78), a recall of 76% (0.76), and an F1-score of 77% (0.77). This means it accurately identified most bad posture instances, but a few cases were misclassified. For good posture, the model performed slightly better, with a precision of 80% (0.80), a recall of 77% (0.77), and an F1-score of 76% (0.76). This shows a fair balance between correct predictions and accurate recognition. The overall accuracy of 77% (0.77) indicates that the model classified postures correctly about three out of four times. The macro and weighted averages also both stood at 77%, reflecting consistent, though moderate, performance in both posture categories. These results show that while the unoptimized model has reasonable classification ability, it lacks balance and stability across different posture types. More optimization is needed to improve its classification consistency and ensure more reliable posture detection.

4.2.2 Optimized Model

Detailed Classification Report (Best Model)	
Model Accuracy	89.71% (0.8971)
Model Loss	29.82% (0.2982)
Model AUC	90.35% (0.9035)

Figure 13. Optimized Metrics

Figure 13 achieved a Model Loss of 29.82% (0.2982), an Accuracy of 89.71% (0.8971), and an AUC of 90.35% (0.9035). The optimized model placed higher probability on correct labels and is less over-confident towards errors, this is further evident with the Accuracy (~9/10 correct at 0.5 threshold) with an AUC that presented ~0.90 across all thresholds, validating strong class discrimination. The researchers cleaned the dataset and confirmed labels through an expert physical therapist, resulting with every sample processed through a consistent pipeline that selected the final model as the checkpoint with the lowest validation loss, prioritizing accurate and well-calibrated predictions toward unseen data. These steps assisted in reducing loss without shifting the decision boundary, yielding a model that is both accurate and reliably confident for real-world ergonomic monitor.

Detailed Classification Report (Best Model)				
	Precision	Recall	F1-Score	Count
Bad Posture	91% (0.91)	88% (0.88)	90% (0.90)	34
Good Posture	89% (0.89)	91% (0.91)	90% (0.90)	34
Accuracy			90% (0.90)	68
Macro Avg	90% (0.90)	90% (0.90)	90% (0.90)	68
Weighted Avg	90% (0.90)	90% (0.90)	90% (0.90)	68

Figure 14. Optimized Class Results

Figure 14 summarizes the model's performance across multiple evaluation metrics for both bad and good posture classifications, providing a comprehensive analysis of the best posture recognition model. The model achieved an overall accuracy of 90% (0.90), indicating that 90% of the postures predicted as either good or bad were correctly classified. For bad posture, the model attained a precision of 91% (0.91) and a recall of 88% (0.88), showing that while it is highly precise in identifying bad posture, it occasionally misses some instances. For good posture, the model achieved a precision of 89% (0.89) and a recall of 91% (0.91), demonstrating a balanced detection capability. The F1-score for both categories is 90% (0.90), reflecting a solid equilibrium between precision and recall. With a total of 68 samples, the model exhibits dependable and consistent classification performance across both classes. The macro and weighted average values of 90% (0.90) for precision and 90% (0.90) for recall and F1-score indicate that the model performs evenly across all categories without significant bias. Overall,

this table highlights the model's strong ability to distinguish between good and bad postures with minimal classification errors and stable predictive performance.

4.2.3 Other Model

This study uses a comparative model developed by Choi et al. [35] in their work, "Hip Positioning and Sitting Posture Recognition Based on Human Sitting Pressure Image," published in *Sensors*. Their model uses a pressure-sensor seat system to determine how people are sitting by analyzing the pressure patterns under them. The image is then analyzed using a hip-template localization method to find three key anatomical regions: the left hip, right hip, and caudal area. After extracting features, a Support Vector Machine with a polynomial kernel is used to classify the data. The dataset contains 800 pressure images from ten people doing four main postures: standing up, leaning back, leaning left, and leaning right. Each posture was done three times with the legs crossed (normal, left-leg-crossed, right-leg-crossed). The SVM classifier reached a maximum accuracy of 89.6%, which shows that it can reliably tell the difference between the four posture categories in controlled settings.

The model works well without visual data because it can detect posture from small changes in pressure. However, it needs special equipment built into the seat, which makes it less portable and more expensive than camera-based options. It also only measures pelvic pressure, so it does not give much information about how the upper body is aligned. Compared to the optimized model used in this study, it highlights the reference system as less accurate and can't be used in different work settings. Both models aim to increase awareness of ergonomics and find people who are sitting in an unhealthy way. However, the model proposed in this paper only works with RGB video input, which allows for non-contact detection and makes it easier to use in office settings. The proposed model also did better than the pressure-based model, with a

classification accuracy of 89.71% and an AUC of 0.9035. This means that it is easier to spot bad postures using upper-body landmarks like the head, shoulders, and spine. These are important parts of ergonomic evaluation for office work.

The current study intentionally focuses on posture assessment to upper-body landmarks, having more data allotted from model training. This methodological choice is supported by recent ergonomic and computer-vision studies, which show that the orientation of the head, neck, shoulders, and spine is the main factor affecting postural health during seated computer work. In contrast, leg positioning has little impact on musculoskeletal risk. Tang et al. showed that posture recognition models that only used upper-body inertial data from the thoracic and lumbar regions were more than 98% accurate. This proves that lower-limb inputs are not necessarily needed to classify upper-body posture correctly. Their recurrent neural network architecture concentrated exclusively on the torso's angular displacement and shoulder-to-hip orientation, determining that the incorporation of leg-sensor data elevated computational expenses without enhancing recognition precision [36]. Lee et al. also stated that bad neck and spine positions, like a forward head, rounded shoulders, and slouching, are the most common ergonomic problems for people who use computers. Their image- and skeleton-based framework attained superior performance by focusing exclusively on upper-body joints (head, neck, shoulders, and spine) and neglecting leg position, due to occlusion and the insignificance of lower limbs in desk settings [37].

Based on these past studies, the current study focuses mostly on upper-body key points (head, neck, shoulders, elbows, and hips) extracted from MediaPipe Pose. This lowers the number of dimensions in the data, speeds up inference, and has higher accuracy. Most office setups put cameras at the height of the desk or monitor, so the legs aren't visible all the time which still makes the model viable. The selected upper-body-focus technique is both ergonomically effective and optimal for real-time monitoring in an office environment.

Comparative Results				
Author / Model	Set Variations	No. of Features	No. of Posture Types	Accuracy
Choi et al. (2021)	Yes	3 (Pressure Zones)	4 (Upright, L/R Lean, Back)	89.6 %
Optimized Model	Yes	29 (Upper-Body Landmarks)	4 (Good, L/R Lean, Slouch)	89.71 %

Figure 15. Comparison Table

Figure 15 shows the compared model included leg variants (normal, left-crossed, right-crossed) to facilitate pressure redistribution across the seat; nonetheless, these configurations mostly reflect weight-shift patterns rather than spinal alignment. Their inclusion enhances the dataset's variety, although it does not provide direct evaluation of ergonomic posture. This study examines upper-body alignment, specifically targeting slouching, forward-head posture, and shoulder asymmetry, which are the principal factors contributing to neck and back discomfort during prolonged sitting. By omitting lower-body inputs, we may concentrate on the biomechanically significant aspects of office ergonomics while maintaining the program's efficiency and user-friendliness. The proposed model demonstrates a 89.71% accuracy and a 0.9035 AUC, indicating its capability to reliably diagnose poor sitting posture only based on visual cues from the upper body. This

corresponds with the study objective of developing a non-invasive, camera-based ergonomic monitoring system that delivers real-time feedback without requiring specialist hardware or full-body visibility.

4.3 Comparative Model Evaluation

The comparison of the improved YOLOv5, MediaPipe, and TensorFlow models, as well as the reference model's pressure-sensor approach, shows that computer vision has considerable benefits in posture identification. With an AUC of 0.9035, a model loss of 29.82%, and an accuracy of 89.71%, the new model successfully categorized postures from RGB images using only upper-body keypoints. In contrast, Choi et al.'s model reached 89.6% accuracy using seat sensor pressure data. These results indicate that the visual method is both effective and superior to hardware-based solutions. The main improvement comes from YOLOv5 for person detection and MediaPipe Pose for extracting upper-body landmarks, which together provide precise identification of shoulder, neck, and spine alignment under a variety of lighting and camera conditions. Because the updated model relies solely on visual input, it is more adaptable to real office environments and does not require specialized equipment or controlled seating. Since the updated model relies only on visual input, it is more adaptable for real office settings, unlike pressure-sensor systems that need special equipment and controlled seating.

The results confirm that upper-body orientation by itself can reliably predict how well someone is sitting at a desk. Previous studies by Tang et al. and Lee et al. show that the alignment of the head, neck, and spine is key for assessing posture. The improved model increases detection accuracy and makes the process easier, since it does not require intrusive or specialized sensors.

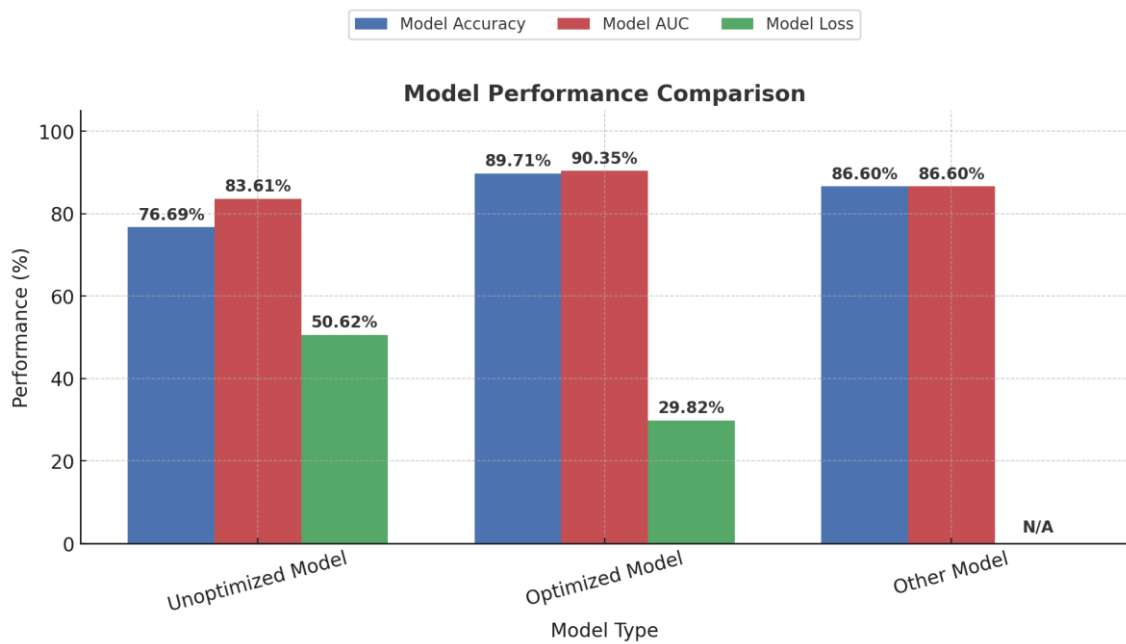
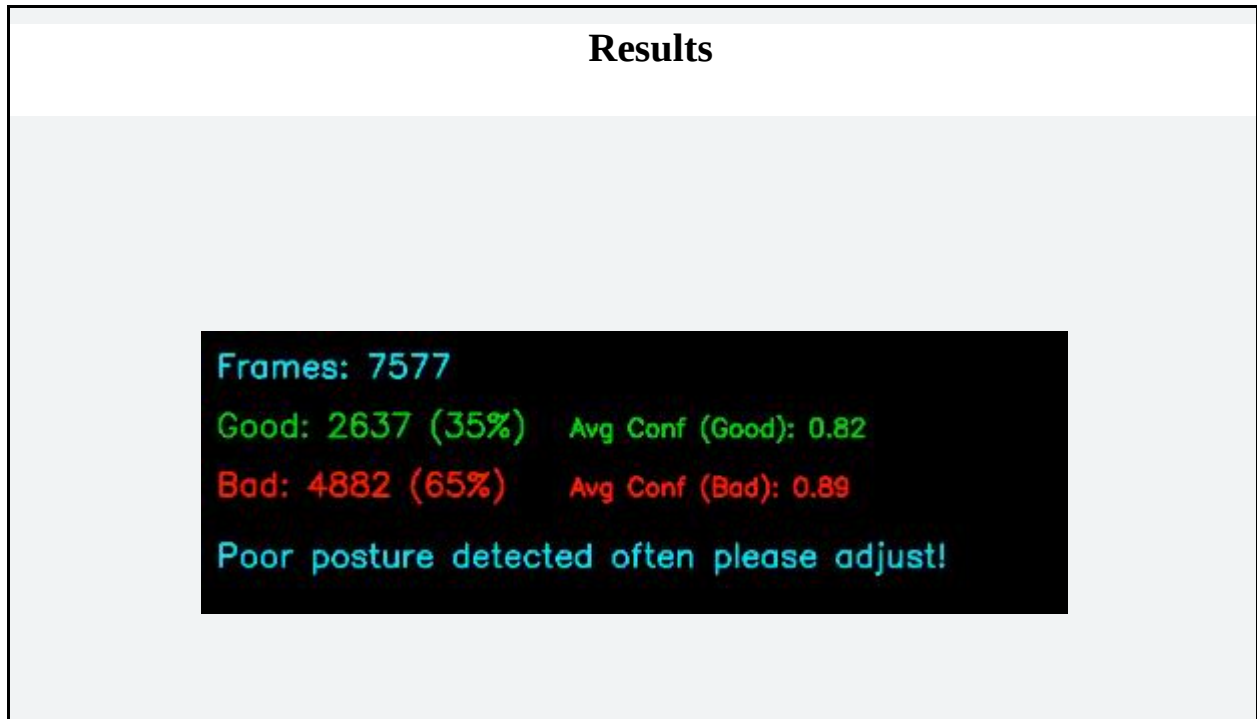


Figure 16. Comparative Result

Figure 16 compares the Unoptimized, Optimized, and Other models, showing the performance gains from model optimization. The Optimized Model had the best results, reaching an accuracy of 89.71% and an AUC of 90.35%. This indicates a strong ability to tell apart good and bad posture classifications with few mistakes. In contrast, the Unoptimized Model had lower scores, achieving 76.69% accuracy and 83.61% AUC. This suggests that its limited parameter tuning and lack of optimization reduced its effectiveness. The Other Model delivered moderate performance, reaching 86.60% in both accuracy and AUC. This shows balanced but slightly lower results compared to the Optimized Model. Regarding loss, the Optimized Model had the lowest at 29.82%, which indicates better learning efficiency and stability. The Unoptimized Model had a higher loss of 50.62%, reflecting weaker convergence. These results confirm that adjusting parameters, improving data processing, and optimizing algorithms significantly enhanced model precision and reliability. Overall, the

Optimized Model is the most consistent and effective setup for posture recognition, balancing high predictive performance with better generalization.

4.4 Model Example Results



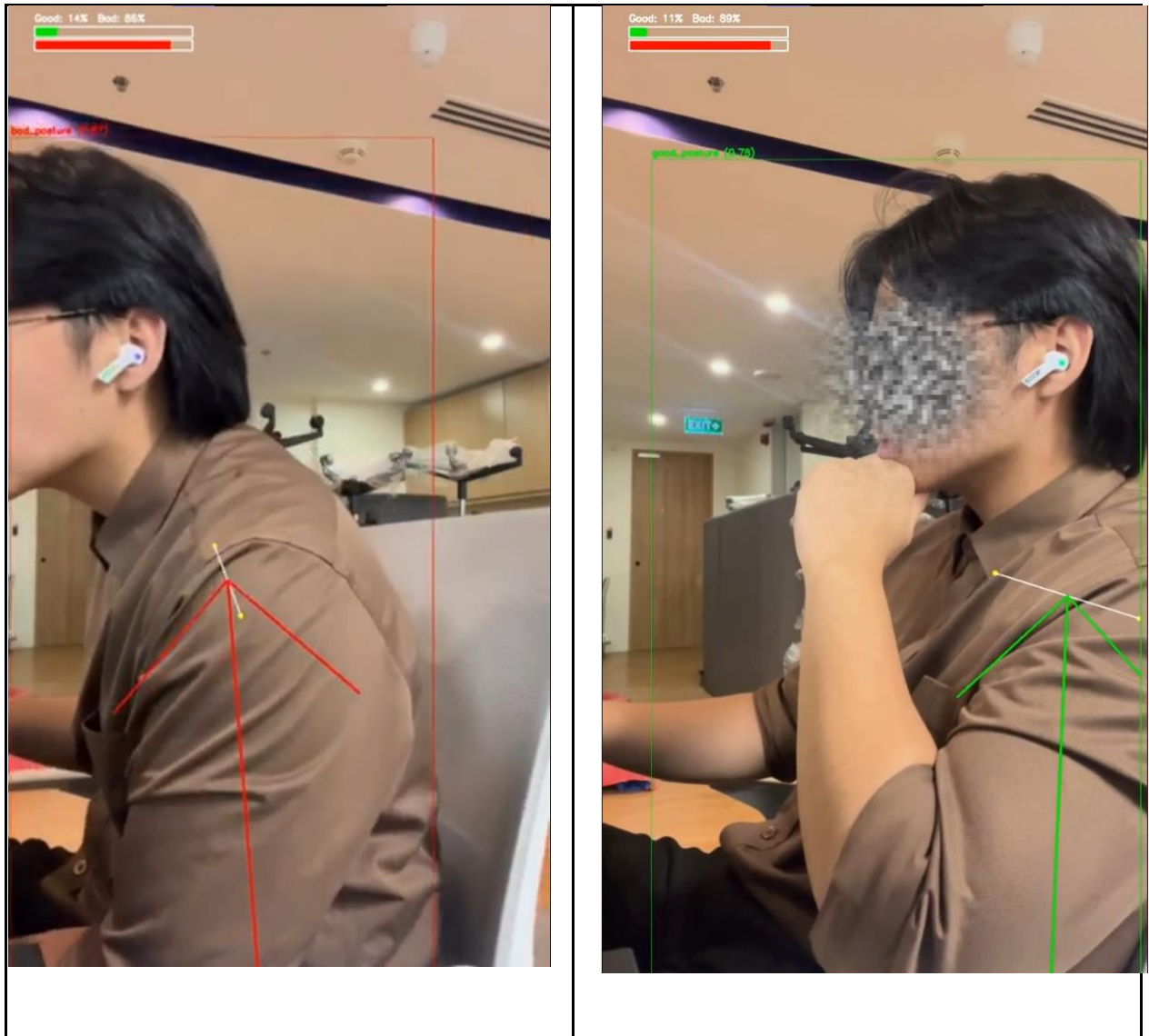


Figure 17. Model Results

Figure 17 shows the experimental results demonstrated that the proposed model effectively utilizes upper-body data for posture detection via camera. The system achieves 89.71% accuracy and an area under the curve (AUC) of 0.9035, supporting its capability to identify postural abnormalities in office environments. The findings have significant practical implications. The proposed non-contact, vision-based system continuously monitors employee posture, addressing limitations associated with wearable devices and physical sensors. This approach aligns with the objective of developing a cost-effective, unobtrusive solution suitable for conventional workspaces using standard RGB cameras. The optimized model demonstrates

superior performance compared to previous studies in both methodology and functionality. While Choi's method required participants to alter their leg position to simulate hip pressure variations, the present model analyzes upper-body landmarks, which are more pertinent to ergonomic assessment. Omitting lower-limb keypoints reduces computational complexity, enhances detection stability under occlusion, and targets postural regions most associated with neck and back pain. The observed increase in accuracy from 89.6% to 89.71% indicates that this upper-body-focused architecture improves performance while preserving interpretability and responsiveness. The application of weighted loss functions stabilized model training and mitigated class imbalance resulting from a limited number of poor posture samples. These modifications facilitated generalization across diverse individuals and environments.

The improved model is more accurate and easier to deploy than the pressure-based technique proposed by Choi et al., according to the comparative evaluation. The results provide validity to the notion that upper-body landmarks alone are sufficient for accurate posture classification and better ergonomic evaluation in the workplace. The model's high accuracy, real-time detection, and hardware independence not only meet the study objectives, but also represent significant improvements above earlier posture-recognition systems. Improved ergonomic safety and productivity in the workplace may be possible with the addition of feedback systems, the ability to identify several users at once, and the ability to track posture over time.

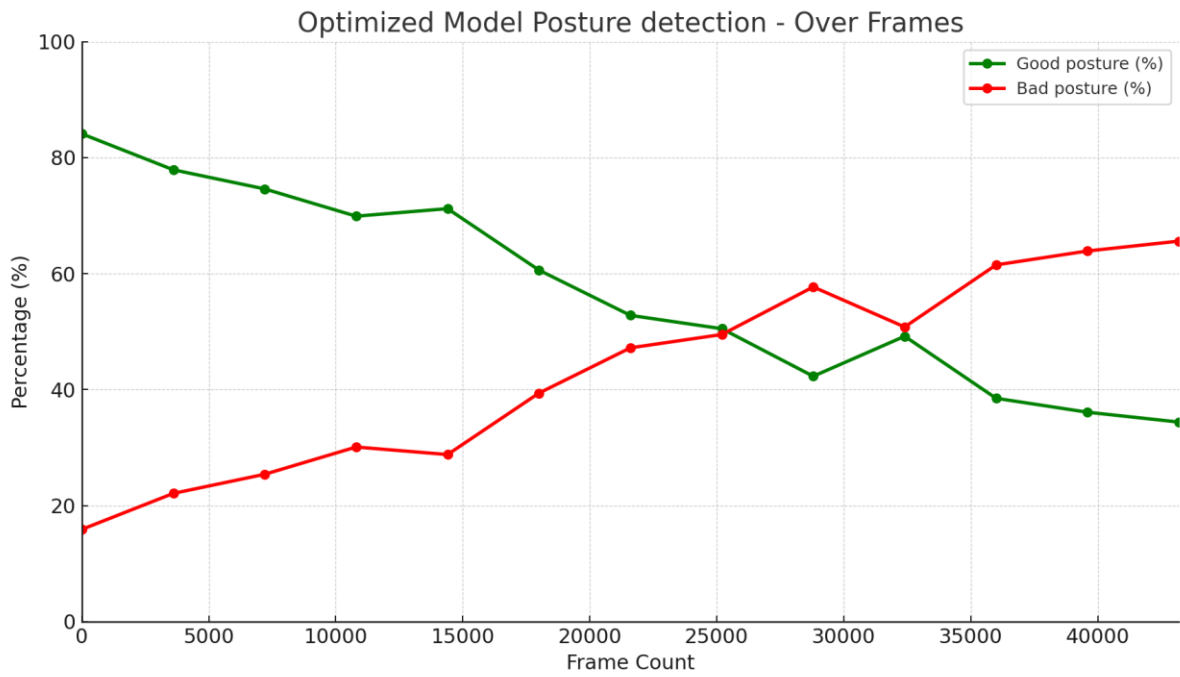


Figure 18. Posture Graph in Frames

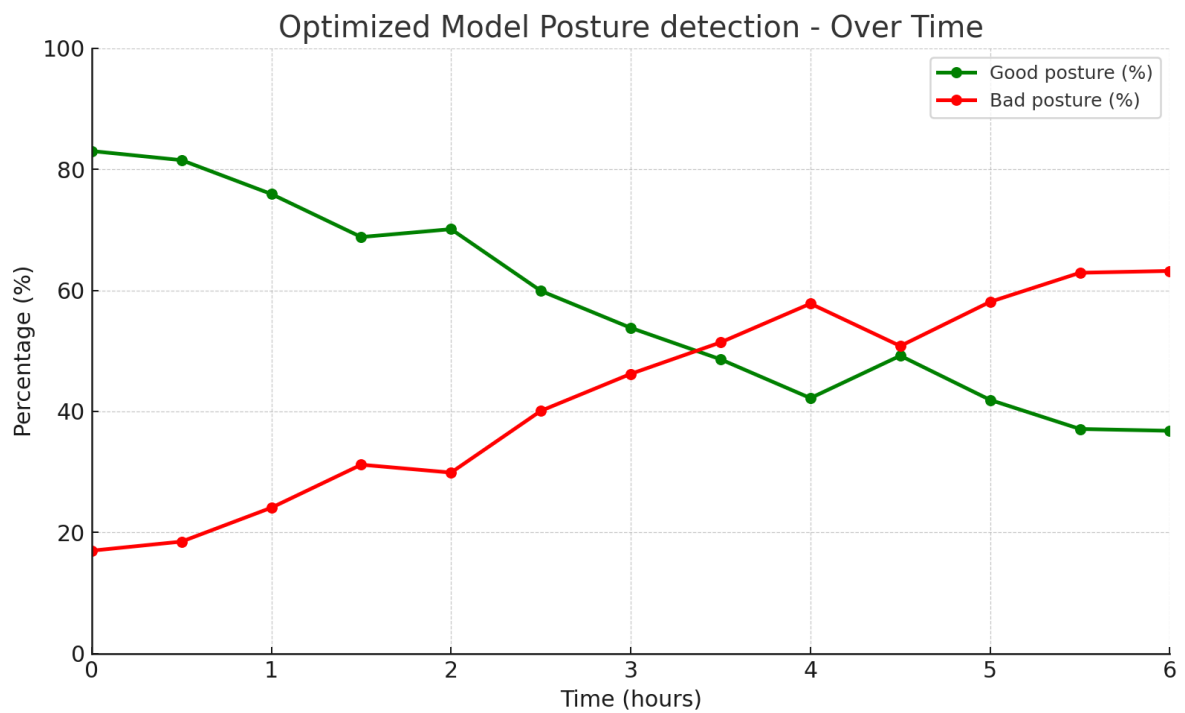


Figure 19. Posture Graph in Hours

Chapter 5

5.1 Overview

The purpose of this research was to establish and evaluate an improved computer vision-based model for workplace seating posture recognition. The objectives, procedures, and key findings of the study are clarified in this chapter. It centers on the fact that the suggested model could accurately and instantly identify incorrect sitting positions by concentrating just on visual cues. The system showed that it is possible to accurately classify posture without using intrusive sensors or specialized hardware by combining YOLOv5, MediaPipe Pose, and TensorFlow. This created a useful and easy-to-use framework for ergonomic monitoring.

5.2 Summary of Findings

This study's primary intention was to design a machine learning system that could classify different sitting positions just by observing the user's upper-body behavior. The study was an effort to focus on the upper body since, according to ergonomic research, good postural health is more significant on how the shoulders, neck, and spine are aligned. The overall goal of this endeavor was to create a model that could be used in regular office settings where cameras are typically placed at a height where the legs are obscured, such as at a desk or monitor. Utilizing YOLOv5 to person detection, MediaPipe Pose to landmark localization, and TensorFlow for object classification, the suggested method attained substantial quantitative results. With a loss value of 29.82% and a classification accuracy of 89.71%, the optimized model obtained an area under the curve (AUC) of 0.9035. Such results demonstrate that the model is stable and efficient during inference while being able to distinguish between good and bad sitting postures. Using specialized gear to detect differences in seat pressure, Choi et al. obtained an accuracy of 89.6%

percent with their pressure-based model. Despite not touching anything and relying solely on vision, this comparison highlights how computer vision solutions can surpass physical sensor systems when it comes to posture recognition.

There was a distinct function for every component of the model in this display. While YOLOv5 allowed for precise person localization in each frame, posture analysis was unaffected by minor body movement. Using MediaPipe Pose, the researchers established a reliable collection of 29 upper-body metrics that displayed important ergonomic features, including the posture of the shoulders, neck, and spine. A contrast to this is TensorFlow, which allows adaptive postural deviation classification through iterative improvement which subsequently simplifies training. All of these pieces came together to form an adequate end-to-end framework that could detect posture in real time using a regular video camera.

Putting focus on upper-body variables in training and evaluation was a major choice in methodology. Research by Tang et al. and Lee et al. lend support to this decision by demonstrating that characteristics of the upper body are sufficient for determining if an individual is sitting properly. In addition to reducing the need for processing resources and eliminating inconsistencies that could arise when legs were obstructed, this strategy made it easier to generalize across diverse lighting and camera situations. This showed that the model could be applied in real-world scenarios and proved that it was beneficial outside of its original setting. The model is also further proven to be helpful in the community as it supports the goals of SDG 3 which promotes good wellbeing and SDG 8 which focuses on promoting safe working environments.

5.3 Discussion of Research Questions

Long static posture and repetitive motions that strain particular muscle groups help extended computer use cause a progressive change in posture. Users frequently slouch, lean forward, or take on asymmetric positions as muscle tiredness sets in over time. This causes spinal misalignment and uneven pressure distribution. This can show up as neck stiffness, back pain, shoulder tension, and poor circulation, which can make a person feel really uncomfortable all day long. Left uncorrected, ongoing exposure to such wrong postures not only compromises current comfort but also causes long-term musculoskeletal problems. Algorithms such MediaPipe and YOLOv5 have demonstrated great ability in spotting wrong seating positions by use of keypoint detection and real-time analysis in response to this problem. MediaPipe is ideal for real-time feedback in regular office settings as it provides quick tracking of body landmarks with little computing needs. YOLOv5, on the other hand, is extremely accurate in identifying and categorizing posture patterns even under diverse lighting and camera angles. With YOLOv5 outperforming in detection accuracy and MediaPipe in speed and accessibility, both methods can successfully provide corrective feedback by categorizing posture deviations and urging users to change. Taken together, these techniques show great promise for encouraging ergonomic awareness and warding off workplace posture-related pain.

5.4 Conclusions

This study's findings confirm that the proposed YOLOv5, MediaPipe, and TensorFlow model successfully achieves the research goals of manually classifying office postures using just vision-based data. Based on the model's results, evaluating ergonomic posture in the majority of workplaces using primarily upper-body data is effective, as it achieves a high degree of accuracy with minimal computing cost. This outcome supports the central hypothesis that a portable, user-friendly model focused on upper body analysis can perform comparably to sensor-based systems.

These findings suggest that computer vision technologies have significant potential to transform health monitoring and workplace ergonomics. Because the model categorizes and provides feedback in real time without requiring physical contact or specialized seating equipment, it is compatible with standard office configurations, including the use of webcams or security cameras.

This form of integration would allow for the continuous, unobtrusive monitoring of employees' posture, which would increase the prevalence of ergonomically sound practices and, prospectively, reduce the incidence of musculoskeletal problems related to prolonged sitting. Additionally, this model's success will lead to future advancements in smart ergonomics. In addition to demonstrating the technical robustness of the proposed system, the results indicate its potential to contribute to healthier and more productive workplaces.

The results indicate that newer computer vision models could take the place of traditional posture-monitoring systems, bringing better reliability, scalability, and lower costs. This change makes posture analysis more accurate and accessible in different environments. Using YOLOv5, MediaPipe, and TensorFlow together also shows that real-time camera-based ergonomic analysis is both practical and effective, which can help improve workplace ergonomics.

Chapter 6

6.1 Recommendation For Practice

Particularly in businesses where employees are prone to extended sitting such as in BPOs or IT sectors the developed posture recognition system may be used in real office environments. Companies can monitor staff posture in real time without violating privacy by including the system into already present computer systems through webcams or CCTV networks. Upon

detecting bad posture, the system can offer soft alerts or visual cues to promote quick correction. Posture data can now be examined over time to find high-risk people or departments, hence enabling focused ergonomic treatments and health initiatives. This realistic application reduces musculoskeletal stress and tiredness over the long run among employees while promoting a health consciousness culture.

6.2 Recommendation For Future Research

Future studies should concentrate on improving the posture recognition system even more by utilizing developments in computer vision, machine learning, and artificial intelligence. Newer architectures like YOLOv7 or YOLOv8 can be used to improve the accuracy, real-time responsiveness, and overall computational efficiency of the current model, which was created using YOLOv5. The ability of the system to generalize across different scenarios will also be strengthened by expanding the dataset to include a wider variety of postures, lighting conditions, and user environments. The incorporation of complementary technologies like edge computing, sensor-based data collection, or Internet of Things (IoT) connectivity could be a future development. These would enable quicker processing and real-time monitoring, guaranteeing that the system can function well even in environments with limited resources.

Additionally, scalability and large-scale deployments across various workplace settings would be made possible by improving the software architecture to support cloud-based data storage and analytics. The system might become a predictive tool with the addition of data analytics modules that can recognize long-term posture patterns, enabling the early identification of possible ergonomic hazards. Last but not least, adding adaptive interfaces that offer tailored ergonomic feedback and suggestions like recommending suitable office furniture or posture correction methods would improve the system's performance beyond straightforward detection to

intelligent support. With these developments, the study can show how AI-driven computer vision systems can be used in real-world scenarios that connect technology, healthcare, and workplace innovation, further contributing to the field of computer science.

6.3 Recommendation For Policy and Implementation

As part of their occupational health programs, businesses and institutions can create ergonomic policies and training courses using posture recognition technology. Rules might comprise frequent ergonomic evaluations utilizing AI-generated posture reports, the installation of posture monitoring devices in high-risk work locations, and regular lectures encouraging proper sitting habits. Employers can also set up incentive-based wellness initiatives in which upholding proper posture helps earn employee rewards or health benefits. Moreover, firms using posture recognition technologies should guarantee compliance with data privacy and consent policies so that ethical standards are kept in the workplace. Organizations can save long-term healthcare expenses and boost employee productivity by including posture awareness in corporate wellness plans.

6.4 Closing Statement

This study shows the great promise of posture detection systems in helping to solve the rising health problems related with stationary work settings. Problems like musculoskeletal disorders, neck strain, and as modern workplaces especially in sectors including IT, BPO, and administration continue to depend significantly on protracted computer use. Chronic back discomfort has become rather widespread. The created system offers an original, non-invasive, and effective answer that not only identifies inappropriate behavior but also harnesses the

strength of machine learning and computer vision. posture also supports real-time behavioral adjustment among staff members. Apart from its primary goal, the system helps to promote ergonomic awareness and change corporate attitudes on occupational health and safety. Implementing such technology can have long-term advantages including lower health-related costs, improved productivity, and less employee tiredness. Significantly, the study emphasizes how artificial intelligence may be employed to advance human well-being by fusing the future's technological development with sincere concern for people's health. The posture recognition system is, in essence, a step to move toward designing environments that are also deeply sensitive to not just being more intelligent and efficient. For the physical and mental health of everyone.

6.5 Acknowledgement

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Appendix



Figure 20. PT Report

Consent Form	
I, REDACTED Hereby give my full consent and agreement to the use of my recorded video data for the research study entitled "Machine Learning-Based Sitting Posture Recognition for Prolonged Use of Technology in Office Environments." I understand that the data collected will be used solely for academic and research purposes, including the demonstration of the model and presentation of the study's results. I acknowledge that I have been informed of the purpose of the study, the scope of data usage, and my rights as a participant. I fully consent to the researchers' use, processing, and analysis of my data in accordance with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173). Furthermore, I understand that my participation is voluntary and that the researchers will ensure the confidentiality, security, and proper disposal of my data after the completion of the study. REDACTED REDACTED 26/10/2025	I, REDACTED Hereby give my full consent and agreement to the use of my recorded video data for the research study entitled "Machine Learning-Based Sitting Posture Recognition for Prolonged Use of Technology in Office Environments." I understand that the data collected will be used solely for academic and research purposes, including the demonstration of the model and presentation of the study's results. I acknowledge that I have been informed of the purpose of the study, the scope of data usage, and my rights as a participant. I fully consent to the researchers' use, processing, and analysis of my data in accordance with the Philippine Data Privacy Act of 2012 (Republic Act No. 10173). Furthermore, I understand that my participation is voluntary and that the researchers will ensure the confidentiality, security, and proper disposal of my data after the completion of the study. REDACTED REDACTED 26/10/2025

Figure 21. Consent Forms